TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Detailed Practical Study of Mechanisms and Operation of a Circular Knitting Machine

The single-jersey circular machine taken apart by eye: cylinder and needles, sinker ring, cam systems, feeders and yarn input, takedown and winding — each mechanism's function stated and the machine's production account calculated from its recorded settings.

FM-4202 · ADVANCED KNITTING SESSIONAL · EXPERIMENT 1 · FABRIC MANUFACTURING SPECIALIZATION

1. Aim and theory

Aim: to identify and explain the working mechanisms of the laboratory circular knitting machine, to record its specification and settings, and to compute its theoretical production.

The knitting cycle (TE-3101) — clearing, yarn feeding, loop formation, knock-over — is executed by the cooperation of four mechanism groups: the **needle cylinder** (needles in tricks, raised and lowered by cams), the **sinker ring** (sinkers holding the fabric down and supporting loop formation), the **feed system** (creel, tensioners, feeders delivering a metered length of yarn per course — the stitch length control), and the **takedown and winding** (drawing the tube down at constant rate and rolling it). On a revolving-cylinder machine the needle set rotates through stationary cam tracks at each feeder; every feeder therefore knits one course per revolution, and production follows from feeders × speed.

2. Apparatus

- Laboratory single-jersey circular machine (stopped for study, then run), tachometer, needle and sinker samples, cam-box section or spare cam set, stitch-length measuring kit (course length method: marked 100-stitch length), GSM cutter and balance, machine manual.

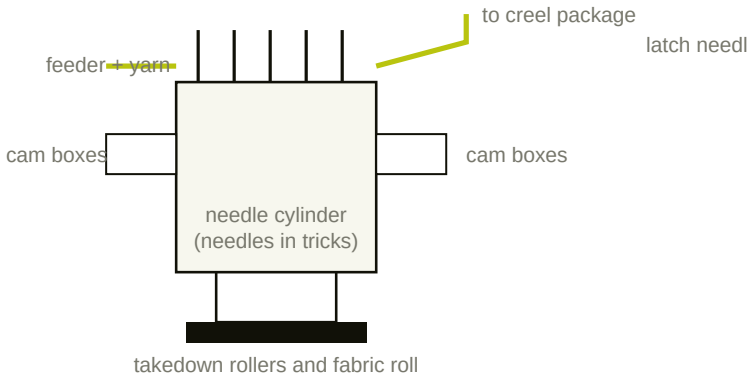


Figure 1 — The circular knitting machine in schematic section: yarn from the creel passes the feeder to needles rising in the cylinder tricks; cam boxes around the cylinder drive the cycle at each feeder; the formed tube descends through the cylinder to the takedown and roll.

3. Procedure

1. **Specification record:** from the machine plate and by measurement — cylinder diameter, gauge, total feeders, feeder type, drive and speed range, takedown type. Compute the needle count and confirm by the plate.
2. **Needle and sinker study:** with the machine stopped, examine a needle: butt, stem, hook, latch and its spoon; trace its path through one cam track by hand-turning — note clearing lift, the latch opening by the old loop, yarn reception, latch closing, knock-over. Examine sinkers: nose, throat and their holding-down role at knock-over.
3. **Cam system:** identify at one feeder the clearing cam, stitch (knock-over depth) cam and guard cams; record the stitch-cam setting — it, with feeder metering, fixes stitch length.
4. **Feed system:** trace the yarn path: package → tension disc → stop sensor → positive feed unit → feeder guide → needles. Record feeder settings (tape/wheel positions or electronic values) per feeder group.
5. **Takedown and winding:** identify the takedown drive and its setting; note how fabric tension is kept and how the roll is driven.
6. **Running study:** run the machine; measure rpm; observe loop formation at one feeder; measure the stitch length delivered (100-stitch course-length method) and the grey CPC/WPC on the tube; collect production data for Section 4.

4. Worked calculation (model data)

PRODUCTION ACCOUNT OF THE STUDY MACHINE

Machine: 30-inch diameter, 24 gauge, 72 feeders, running 28 rpm. Needles = π × 30 × 24 ≈ **2262**. Courses per minute = feeders × rpm = 72 × 28 = **2016 courses/min**.

At grey CPC 16/cm: tube length rate = 2016 ÷ 16 = 126 cm/min = **75.6 m/h** of tube.
Mass rate at GSM 140 g/m² and tube circumference π × 30 in = 2.39 m: 75.6 × 2.39 × 140 ÷ 1000 ≈ **25.3 kg/h** at 100% — the ceiling against which the shift's actual output (stops for faults, doffing, yarn changes) is judged.

5. Observation table (as recorded — model data)

ITEM	RECORD
Cylinder / gauge / needles	30 in / 24 gg / 2262 (computed, plate agrees)
Feeders	72, positive feed, tape drive
Speed	28 rpm (tachometer)
Stitch length delivered	3.2 mm (course-length method, feeder group average)
Grey geometry	CPC 16/cm, WPC 11/cm on the tube (dry state, on machine)
Takedown	Roller takedown, setting per quality sheet

6. Precautions

- Hand-turn only in the marked direction, with guards respected; never reach across the rotating cylinder or into the feeder zone while running.
- Loose clothing, jewellery and hair secured before any running observation — the cylinder forgives nothing.
- Stitch-length measurements are taken on yarn drawn from the running machine by the prescribed method — never by unravelling a moving tube.

7. Practice (viva preparation)

1. State the separate functions of needle, sinker and stitch cam in one loop-forming cycle.
2. A 26-inch, 18-gauge machine: compute its needle count.
3. The machine of Section 4 is re-set to 24 rpm with the same feeders and fabric. Recompute its mass rate.
4. Why does production follow feeders × rpm, and what machine family breaks that rule?
5. What single measurement integrates feeder settings across the whole machine?

Answer key

1. **Needle:** rises to clear the old loop below the latch, receives new yarn, descends drawing it through the old loop. **Sinker:** holds the fabric/old loop down so the needle can rise through it and supports the yarn at knock-over. **Stitch cam:** sets how far the needle descends — hence how much yarn is drawn, i.e. the stitch length.
2. $\pi \times 26 \times 18 \approx$ **1470 needles**.
3. Courses/min = 72 × 24 = 1728; length = 1728 ÷ 16 = 108 cm/min = 64.8 m/h; mass = 64.8 × 2.39 × 140 ÷ 1000 ≈ **21.7 kg/h**.
4. Each feeder knits one course per cylinder revolution, so courses accumulate feeders-per-revolution × revolutions. **Flat machines** break the rule: courses are formed per carriage traverse, with systems per carriage instead of feeders per revolution.
5. The **delivered stitch length** (course-length measurement): it is the common output of every feeder's metering and the cam settings together, and it alone predicts the fabric's relaxed geometry (FM-4202 Experiment 5).

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